Practical Section 2
1D UCLCHEM

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Parameter setup

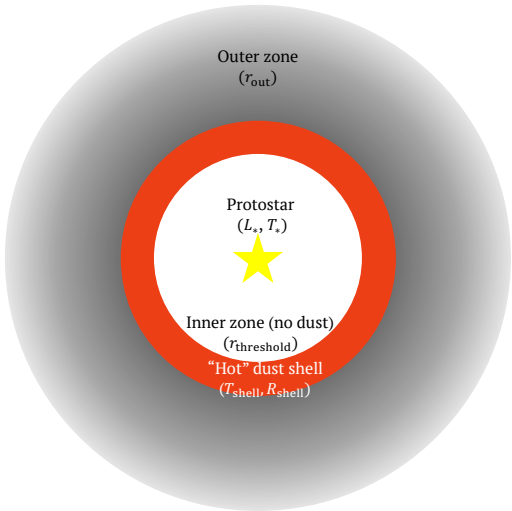

Figure 1: The sketch of the protostellar core adopted in this model, including a central source characterised by the bolometric luminosity (L_*) and the stellar temperature (T_*). The inner zone is characterised by the sublimation threshold of dust grains ($T_{\rm sub}$) and the threshold radius ($r_{\rm threshold}$). The thin shell of hot dust is determined by the shell temperature ($T_{\rm shell}$) and its radius ($R_{\rm shell}$). The outer region is given by the outer radius ($r_{\rm out}$).

The gas volume density at a distance, r, in the cores is adopted in the Bonnor-Ebert sphere as

$$n_{\rm gas}(r) = \begin{cases} n_0 & \text{for } r \leq r_0, \\ n_0 \left(\frac{r}{r_0}\right)^{-\alpha} & \text{for } r > r_0, \end{cases} \tag{1}$$

where n_0 is the maximum density in the centre and is a parameter of our model, along with α and r_0.

The input parameters:

- Bolometric luminosity and temperature (can be adopted from the HR diagram)
- The peak gas volume density (n_0), r_0 and the slope α

Hint:

- *Catalogs with $L_* = 1,\ 10^1,\ 10^2,\ 10^3,\ 10^4,\ 10^5,\ 10^6\, L_\odot$ are given (generated by [1])*
  - *Respectively, $r_{\rm out} = 0.01,\ 0.015,\ 0.03,\ 0.06,\ 0.1,\ 0.2,\ 0.5\,$pc*
  - *Respectively, $n_0 = 10^8,\ 10^8,\ 10^8,\ 10^7,\ 10^7,\ 10^7,\ 10^7\,{\rm cm}^{-3}$*
  - $\alpha = 2.4$
  - $a = 0.5\,\mu m$
- *These catalogs can be used directly for your observations (reference can be found in [1])*

Exercises

1. Visualize the physical properties (gas volume density, gas temperature, visual extinction, ...)

   Hint:

   - *100 points are sampled uniformly between* $r_{\rm in}$ *and* $r_{\rm out}$ *(with* $r_{\rm in} = r_{\rm out}/100$*)*
   - *the first line of the data file should be ignored (it corresponds to the initialization)*
   - *the data block at point=100 corresponds to* $r_{\rm out}$

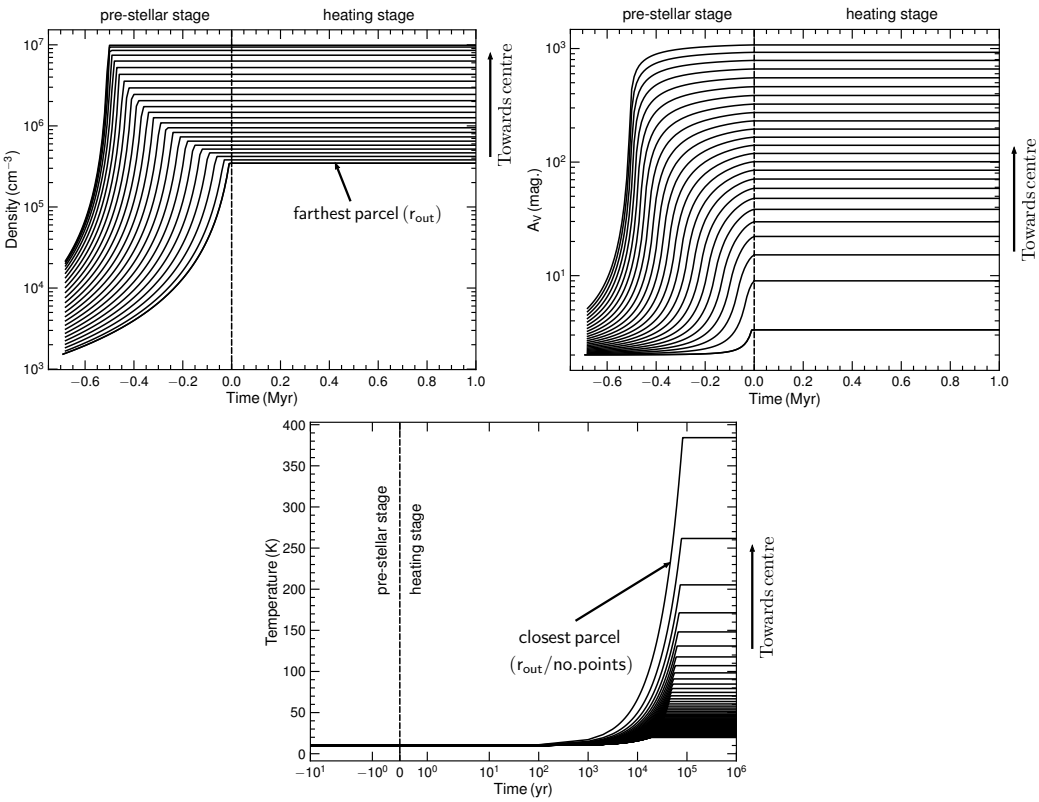

Figure 2: An examples of the physical profiles.

2. Visualize/investigate the chemical properties of the molecules of interest.

   Hint:

References

1. Tram, L. N. *et al.* One-dimensional and time-dependent modelling of complex organic molecules in protostars. *arXiv e-prints,* arXiv:2601.00731. arXiv: 2601.00731 [astro-ph.GA] (Jan. 2026).

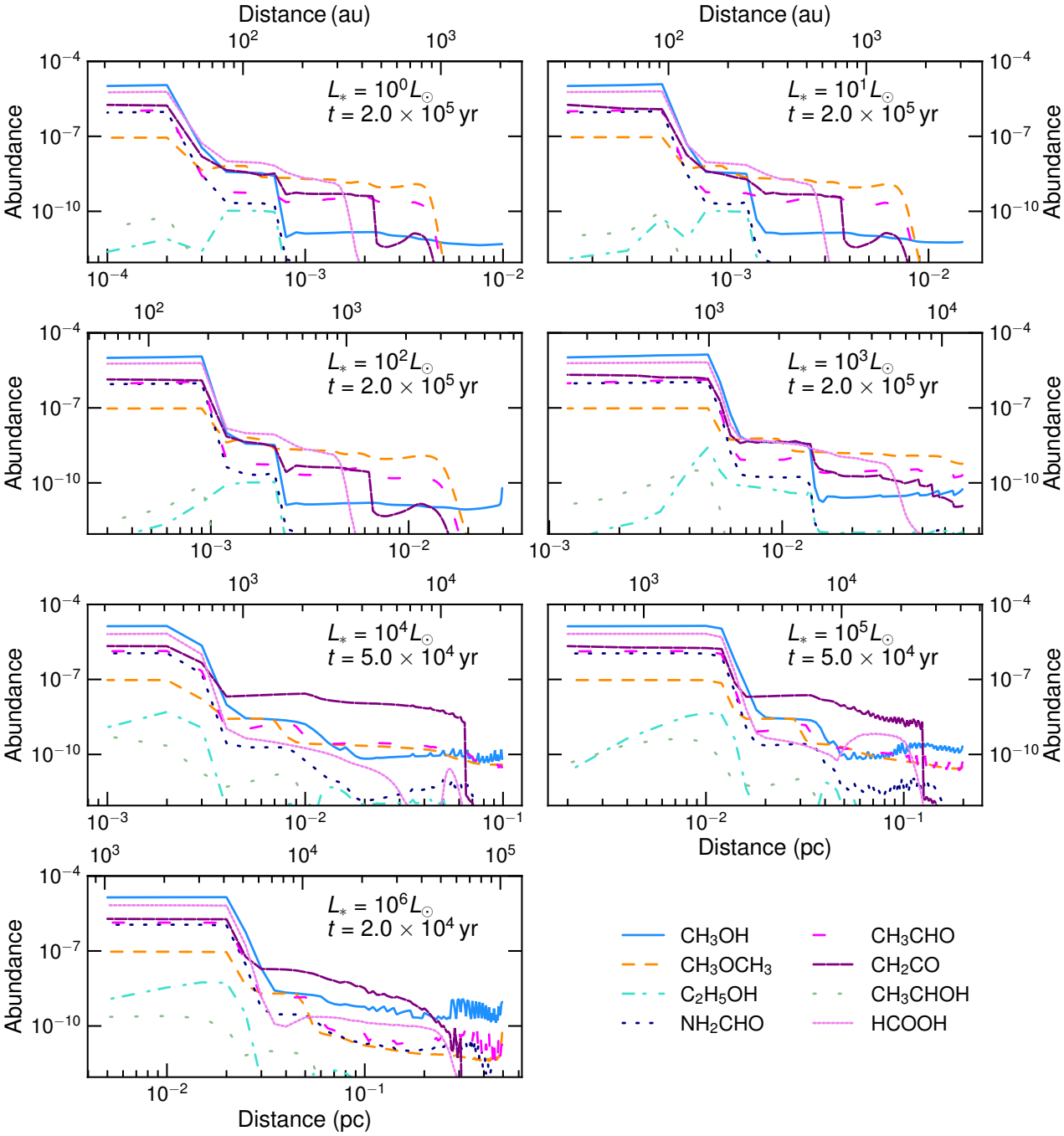

Figure 3: An example of the chemical profiles.